

Exercise 2.8. Prove the following. For each, m , n and t are integers.

(a) $1 \mid n$.

(c) If $mn \mid t$, then $m \mid t$.

(b) $n \mid n$.

(d) If $mn \mid tn$, then $m \mid t$.

Solution.

Sure. We use the definition of divisibility throughout:

$$a \mid b \iff b = ak$$

for some integer k .

(a) $1 \mid n$

We need to show that $n = 1k$ for some integer k .

Take $k = n$. Since n is an integer,

$$n = 1 \cdot n.$$

Therefore,

$$\boxed{1 \mid n}.$$

(b) $n \mid n$

We need to show that $n = nk$ for some integer k .

Take $k = 1$. Then

$$n = n \cdot 1.$$

Since 1 is an integer,

$$\boxed{n \mid n}.$$

(c) If $mn \mid t$, then $m \mid t$

Since

$$mn \mid t,$$

by definition there is an integer k such that

$$t = mnk.$$

Rewrite this as

$$t = m(nk).$$

Since n and k are integers, nk is also an integer. Therefore, t is m times an integer.

Hence,

$$\boxed{m \mid t}.$$

(d) If $mn \mid tn$, then $m \mid t$

Since

$$mn \mid tn,$$

there is an integer k such that

$$tn = mnk.$$

If $n \neq 0$, divide both sides by n :

$$t = mk.$$

Since k is an integer, this means

$$\boxed{m \mid t}.$$

Important: As stated, part (d) needs the assumption $n \neq 0$. If $n = 0$, then $mn = tn = 0$, so the hypothesis $mn \mid tn$ is true for every m, t , but $m \mid t$ need not be true.

So the precise statement is:

$$\boxed{\text{If } mn \mid tn \text{ and } n \neq 0, \text{ then } m \mid t.}$$